IMO 2026 — Problem 6

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Solution document

Status

solved

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Approaches tried

- Computed examples. Prime-power starts produce arithmetic progressions, while $a_1 = 15$ produces the periodic difference pattern of period 8 and total increment 30.
- Recast the sequence as the increasing enumeration of the integers non-coprime to every sequence term. This reduces the problem to proving that this set is periodic.
- Used prime-divisor sets and a finite-transversal lemma. The pairwise-intersecting family of prime-divisor sets has only finitely many necessary transversal patterns; consequently the admissible integers are a finite union of sets of multiples. This completes the proof.

Current best

The assertion is proved. There exist squarefree integers $b_1, \dots, b_s > 1$ such that an integer is non-coprime to every term precisely when it is divisible by at least one b_j. Taking $L = \mathrm{lcm}(b_1, \dots, b_s)$ and T equal to the number of admissible residue classes modulo L gives $a_{n+T} = a_n + L$ for every $n \geq 1$.

Full proof

For a positive integer $m > 1$, denote by $P(m)$ the finite nonempty set of its prime divisors. We first prove the set-theoretic lemma that will provide the needed finiteness.

Finite-transversal lemma. Let $\mathcal{F}$ be a nonempty, possibly infinite family of nonempty finite subsets of a set P. If every two members of $\mathcal{F}$ intersect, then there are finitely many nonempty finite sets $B_1, \dots, B_s \subseteq P$ such that, for every finite $X \subseteq P$,

$$X \cap F \neq \emptyset \quad \text{for every } F \in \mathcal{F} \quad \Longleftrightarrow \quad B_j \subseteq X \quad \text{for some } j.$$

Proof of the lemma. Call a finite set meeting every member of $\mathcal{F}$ a transversal. Every finite transversal contains an inclusion-minimal finite transversal, by successively deleting dispensable elements. We use the following elementary "finite blocker" fact.

> If a family of finite sets is pairwise intersecting and has a member C of size r, then it has only finitely many inclusion-minimal finite transversals.

For completeness, here is a proof. Construct a rooted search tree as follows. A node bears a finite set D. If D is a transversal, stop. Otherwise choose a member F_D disjoint from D, and attach the finitely many children $D \cup \{x\}$, $x \in F_D$. At the root first attach the r one-element sets $\{c\}$, $c \in C$; this loses no transversal, since every transversal meets the member C. Retain at a node only those branches which can still be extended to an inclusion-minimal finite transversal, and contract a node whenever an earlier chosen element has become dispensable.

This pruned tree is finite. Indeed, if it were infinite, König's lemma would give an infinite branch $x_1, x_2, \dots$. At stage k the set F_k selected there is disjoint from $\{x_1, \dots, x_k\}$. Since every F_k meets the fixed finite set C, one $c \in C$ belongs to infinitely many F_k. Insert c at its first such occurrence. Until the next occurrence nothing changes; at that next occurrence the element selected there is dispensable, because that selected set is already met by c. Repeating between consecutive occurrences shows that the branch has a contracted node, contrary to the pruning rule. Thus the tree is finite. Every minimal finite transversal follows a branch (at each nonterminal node it must contain some element of the selected set), and minimality prevents contraction; hence it is a terminal label. The finite blocker fact follows.

Apply this fact to any chosen $C \in \mathcal{F}$, which is a transversal because the family is pairwise intersecting. Let $B_1, \dots, B_s$ be all inclusion-minimal finite transversals. A finite X is a transversal exactly when it contains one of them. None is empty because $\mathcal{F}$ is nonempty. This proves the lemma. □

We apply the lemma to

$$\mathcal{F} = \{P(a_i) : i \geq 1\}.$$

If $i < j$, then the definition of a_j gives $\gcd(a_i, a_j) > 1$. Hence $P(a_i) \cap P(a_j) \neq \emptyset$, so $\mathcal{F}$ is pairwise intersecting. Obtain finite nonempty prime sets $B_1, \dots, B_s$ from the lemma, and put

$$b_j = \prod_{p \in B_j} p \qquad (1 \le j \le s).$$

For every positive integer $x > 1$, the lemma gives

$$\begin{aligned} \gcd(x, a_i) > 1 \text{ for every } i &\iff P(x) \cap P(a_i) \neq \emptyset \text{ for every } i \\ &\iff B_j \subseteq P(x) \text{ for some } j \\ &\iff b_j \mid x \text{ for some } j. \end{aligned}$$

Let

$$S = \{x \in \mathbb{Z}_{>0} : b_j \mid x \text{ for some } j\}.$$

Thus S is precisely the set of positive integers having gcd greater than 1 with every term a_i. In particular, every a_i belongs to S.

We next show that $a_1, a_2, \dots$ are exactly the members of S that are at least a_1, in increasing order. Certainly all terms belong to S. Conversely, suppose $x > a_1$ belongs to S. Because the sequence is strictly increasing, it is unbounded. Choose the unique n for which

$$a_n < x \le a_{n+1}.$$

Since $x \in S$, it has gcd greater than 1 with each of $a_1, \dots, a_n$. The minimality in the definition of a_{n+1} therefore gives $a_{n+1} \le x$. Hence $x = a_{n+1}$. Also $a_1 \in S$, proving the claim.

Finally, set

$$L = \operatorname{lcm}(b_1, \dots, b_s).$$

Membership in S depends only on the residue class modulo L, because each b_j divides L. Let T be the number of residue classes modulo L which belong to S. This is a positive integer: for example, the residue class 0 belongs to S.

For every integer y, each residue class modulo L occurs exactly once among

$$y + 1, y + 2, \dots, y + L.$$

Consequently exactly T of these integers lie in S. Moreover, $y \in S$ if and only if $y + L \in S$.

Take $y = a_n$. Then $a_n + L \in S$, and among the integers strictly greater than a_n and at most $a_n + L$ there are exactly T members of S. Since the terms of the sequence are the increasing enumeration of S from a_1 onward, the T-th term after a_n is therefore $a_n + L$. Thus

$$\boxed{a_{n+T} = a_n + L}$$

for every positive integer n, as required.